

DM Small Group 6 Integrated Lab – Abdominal Pain

By Dr. Mohamed Idries

SOM. MK.I.BPM2.3.DM.2.BCHM.1210	Describe the major steps of cholesterol synthesis, including the role of acetyl CoA as the initiating substrate.
SOM. MK.I.BPM2.3.DM.2.BCHM.1214	Describe the pathway of bile acid synthesis from cholesterol, its regulation, and the formation and differentiation of primary and secondary bile salts.
SOM. MK.I.BPM2.3.DM.3.BCHM.1242	Explain the biochemical steps involved in lipid digestion and absorption, emphasizing the role and recycling (enterohepatic circulation) of bile salts in these processes.
SOM. MK.I.BPM2.3.DM.3.BCHM.1253	Discuss the composition of bile, factors contributing to gallstone formation (cholelithiasis), and predict the consequences of bile duct obstruction on lipid digestion and nutrient absorption.
SOM. MK.I.BPM2.3.DM.3.BCHM.1392	Describe the structure and function of the inner mitochondrial membrane in the electron transport chain, identify major electron carriers in each ETC complex, and trace the path of electrons from NADH and FADH ₂ to oxygen.
SOM. MK.I.BPM2.3.DM.2.BCHM.1393	Describe the biochemical pathway of heme biosynthesis, including the required coenzymes, and explain the regulation of heme production in different tissues.
SOM. MK.I.BPM2.3.DM.2.BCHM.1394	Differentiate between acute intermittent porphyria, porphyria cutanea tarda, and congenital erythropoietic porphyria based on clinical features and laboratory findings, explaining the biochemical basis of each.
SOM. MK.I.BPM2.3.DM.2.BCHM.1395	Discuss how lead poisoning, pyridoxine (vitamin B6) deficiency, iron deficiency, and isoniazid therapy disrupt heme biosynthesis, and explain the underlying metabolic mechanisms.
Physiology Objectives:	
SOM.MK.II.BPM2.3.DM.1.PHYS.1412	Compare and contrast the regulation of the gastrointestinal functions by neural, hormonal, and paracrine regulators.
SOM.MK.II.BPM2.3.DM.1.PHYS.1413	Differentiate major types of gastrointestinal motility patterns with respect to their site of origin, mechanism, and purpose in the gut.
SOM.MK.II.BPM2.3.DM.1.PHYS.1414	Examine the role of intrinsic (enteric nervous system) and extrinsic (sympathetic and parasympathetic nervous system) neuronal regulations of the gastrointestinal motility.
SOM.MK.II.BPM2.3.DM.1.PHYS.1415	Distinguish origin, integration, regulation, and purpose of various gastrointestinal reflexes.
Histology Objectives:	
SOM.MK.I.BPM2.3.DM.1.HCB.1318	Discuss the flow of bile from the hepatocytes where it is made to its exit in the duodenum
SOM.MK.I.BPM2.3.DM.1.HCB.1319	Microscopically describe the cholangiocytes
SOM.MK.I.BPM2.3.DM.1.HCB.1320	Identify the parts of the biliary tree lined by cholangiocytes
SOM.MK.I.BPM2.3.DM.1.HCB.1321	Discuss the function(s) of cholangiocytes
SOM.MK.I.BPM2.3.DM.1.HCB.1322	Discuss the hormonal impact on the function of cholangiocytes
SOM.PB.I.BPM2.5.DM.5.HCB.1323	Generating a deeper understanding of professional and presentation skills
Anatomy Objectives:	
SOM.MK.1.BPM2.3.DM.1.ANAT.1167	Describe the location of the gall bladder and its relationship to the liver.
SOM.MK.1.BPM2.3.DM.1.ANAT.1170	List the 3 common sites where gall stones can be impacted (lodged) and explain the effects of each on the bile flow.
SOM.MK.1.BPM2.3.DM.1.ANAT.1171	Explain how the visceral pain from an inflamed gallbladder can refer to the shoulder.

Case 1:

A 58-year-old woman is brought to the Emergency Department because of a 1-day history of abdominal pain, vomiting and a low-grade fever. The pain is colicky in nature, intermittent, poorly localized, and associated with constipation.

She reports that she has had recurrent episodes of abdominal pain associated with nausea and vomiting for the past 3 days. The pain was severe and steady and lasted for about 30 minutes. The pain is localized to the epigastric or sometimes, right upper quadrant and radiates to the shoulder. The pain occurred after fatty food ingestion and the episodes became progressively worse over the past 3 days.

Her temperature is 37.5°C (99.5°F), pulse is 104/min, blood pressure is 126/84 mm Hg, and respirations are 16/min. Physical examination shows dry tongue and mouth, moderately distended abdomen with increased bowel sounds, and a positive Murphy's sign (Physician elicits tenderness by placing the hand below the right costal margin at midclavicular line while patient takes a deep breath). Plain abdominal x-ray shows dilated small bowel loops with air-fluid levels, absent air in the colon, and air in the biliary tree (pneumobilia). She is diagnosed with acute cholecystitis with gallstone ileus.

She undergoes enterolithotomy (a surgical procedure to create an opening proximal to the point of stone impaction) with stone extraction followed by cholecystectomy (a surgical procedure that involves removal of the gallbladder) and biliary-enteric fistula closure. The gallstone is sent to the laboratory and is shown to be a cholesterol stone. Genetic studies show a loss-of-function mutation in the CYP7A1 gene that encodes for cholesterol 7- α -hydroxylase.

Q1) In the initial episodes why did she experience pain over the epigastric and right upper quadrant?

Q2) Why was the pain felt over her right shoulder?

Q3) Why did she feel sharp pain in the right upper quadrant of her abdomen during abdominal examination? (Murphy's sign)

Q4) Relate the initiation of pain episodes after a fatty meal ingestion. Discuss secretion and functions of CCK.

Q5) List major neurocrine, endocrine and paracrine regulators of GIT. Discuss the cell source, stimuli for release, target cells and key role of endocrine peptides in regulating the GI functions.

Q6) Explain different types of enteric motility? How are they generated and regulated? Which type is responsible for the poorly localized, colicky pain?

Q7) How is the mutation in cholesterol 7- α -hydroxylase related to her clinical presentation?

- **How are bile and bile acids/salts formed? Why are bile salts conjugated? Identify the cells that are responsible for bile production on the histologic slide.**

Which factors play a role in the formation of gallstones?

Q8) What is the function of bile? What are the constituents of bile?

Q9) How does bile flow from hepatocytes into the duodenum?

- What is the function of the cholangiocytes?
- Which hormones influence the function of cholangiocytes?
- Identify the following histologic slide and describe their characteristic features.

Q10) What is the difference between primary and secondary bile acids? Which bile acid would you expect to be predominant in feces?

Q11) Why is enterohepatic circulation important?

- What are all the factors that contributed to this patient's development of gallstones?

As part of her long-term therapy she is asked to be on a diet low in lipids and especially cholesterol, and she is put on a statin.

Q12) What is the source and what are functions of cholesterol? How is cholesterol synthesized and how is the synthesis regulated?

Q13) How will statins therapy result in lowering her plasma cholesterol? Show the changes on a Lineweaver-Burke plot.

She started experiencing muscle pain after two weeks from starting her treatment

Q14) How could her muscle ache be related to her statin treatment?

- Discuss the intermediates of cholesterol synthesis pathway needed for the electron transport chain.

Q15) Where would you expect the surgeon to locate the biliary-enteric fistula?

Q16) During the laparoscopic cholecystectomy (removal of the gallbladder), which artery should the surgeon ligate and how will the surgeon locate it?

Case 2:

Magna Rose is a 19-year-old female who was brought to their family physician by her parents with complaints of recurrent abdominal pain. This was her 5th visit in 6 months. There is a family history of Porphyria. Routine blood tests had always been normal except for mild anemia. On physical examination, she was found to be agitated, confused, with severe abdominal pain and respiratory distress. Limb weakness was also noted. She was given intravenous fluids and pain treatment and was discharged the following day as her condition improved remarkably. Two days later, she reappeared to her physician as she was concerned about the dark discoloration of her urine on exposure to air. Blood and urine were collected for investigation and the results were suggestive of acute intermittent Porphyria. She was subsequently managed based on her new diagnosis and the family counseled. It is important to not prescribe barbiturates for AIP patients, as this would stimulate even more incomplete heme synthesis and dangerous accumulation of ALA and porphobilinogen.

Q1) List features in the history that are suggestive of the diagnosis.

Q2) What is the biochemical defect in Acute Intermittent Porphyria (AIP)?

- List the substrates that accumulate in the blood and urine in AIP.

Q3) What is the treatment of AIP? What is the biochemical basis behind the treatment of this disorder?

- Provide a biochemical explanation as to why barbiturates are avoided in AIP patients?

Q4) Describe the steps involved in heme synthesis. What are the organs/tissues mainly involved?

Q5) Describe the regulation of heme synthesis in the:

- a. Liver
- b. Bone marrow

Q6) Discuss the various types of porphyrias of Acute Intermittent Porphyria (AIP), Congenital Erythroid Porphyria (CEP) and Porphyria Cutanea Tarda (PCT).

- List the enzyme that is deficient, the tissues mainly involved and the compounds that accumulate in blood and urine.

Q7) Describe the action of the accumulated compounds and describe the color of urine upon release.

Q8) Explain the biochemical basis for the presence of photosensitivity in congenital erythropoietic Porphyria (CEP) but not in AIP.

Q9) Explain the biochemical basis for anemia in 1] Lead poisoning and 2] Vitamin B6 deficiency.

- Which compounds accumulate in blood and urine, respectively?
- How can you differentiate lead poisoning from AIP?

Q10) Why was there a color change in this patient's urine upon standing?